

D 110236

(Pages : 2)

Name.....

Reg. No.....

FIFTH SEMESTER (CBCSS-UG) DEGREE EXAMINATION, NOVEMBER 2024

Physics/Applied Physics

PHY 5B 08/APH 5B 08—OPTICS

(2019 Admission onwards)

Time : Two Hours

Maximum : 60 Marks

*The symbols used in this question paper have their usual meanings***Section A — Short Answer type.***Answer **all** questions in two or three sentences,
each correct answer carries a maximum of 2 marks.*

1. Define a thin lens.
2. Define first principal focus of a lens.
3. Define lateral magnification.
4. Explain interference.
5. Explain coherence.
6. Define Fiber Bragg Gratings.
7. Explain colours of thin film.
8. Write down the equation for resolving power of a grating.
9. Define Huygens's explanation of double refraction.
10. Define polarization.
11. What is hologram ?
12. Give two applications of holography .

(Ceiling 20)

Section B - Paragraph / Problem type*Answer **all** questions in a paragraph of about half a page to one page,
each correct answer carries a maximum of 5 marks.*

13. Explain the interference by a plane parallel film when illuminated by a plane wave.
14. Derive cosine law.
15. Write a note on non reflecting films.

Turn over

16. With necessary figure, explain interference by a plane parallel film illuminated by a point source.
17. Derive the equation for width of principal maxima of an N slit Fraunhofer diffraction.
18. Define double refraction. Explain Laurent's half-shade polarimeter.
19. With figure explain Fresnel's half period zone.

(Ceiling 30)

Section C - Essay type

*Essays - Answer in about two pages, any **one** question.
The question carries 10 marks.*

20. Briefly derive and the intensity distribution of interference pattern
21. Explain with figure, single slit diffraction pattern. Explain the positions of maxima and minima
(1 × 10 = 10 marks)